# Voice Assistant - Submission

Real-time offline audio-in / audio-out conversational AI assistant

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## Deliverables (Google Drive links)

**1. Source code (zip) - full project: assistant/ package, tests/, docs, requirements.txt**
https://drive.google.com/file/d/10h0aJThLRWcaVEHrUrfY_jfthubj-1EY/view?usp=drive_link

**2. README.md - setup, usage, features**
https://drive.google.com/file/d/1nlqZqV9qtheEZ1zQodQQhf0H9eFGtzMB/view?usp=drive_link

**3. ARCHITECTURE.md - design, pipeline stages, measured latency budget**
https://drive.google.com/file/d/1snSGnH6pgDazg_zQHY-XGHj-FhaWdzWC/view?usp=drive_link

**4. AI_USE.md - AI use disclosure**
https://drive.google.com/file/d/16bxxsCFHpeizxpvmDO5WEGHWr3M2i-_L/view?usp=drive_link

**5. Drive folder containing all deliverables (including this document)**
https://drive.google.com/drive/folders/14L_IrSzp7HAbx99RALnhjQVHNQFntlMm

## AI Use Disclosure

AI was used in writing the code of this project. All Python modules in the assistant/ package (audio capture, voice activity detection, speech-to-text, LLM integration, text-to-speech, fallback engine, barge-in) and the tests were written by Anthropic's Claude (via the Claude Code tool), working under the author's direction, phase by phase. AI was not used for: project requirements and acceptance criteria, final live microphone testing and acceptance of the working system, or the decision of which architecture trade-offs to accept. Full details in AI_USE.md (link above).

## Solution summary

Fully offline pipeline: microphone -> Silero VAD (end-of-speech detection) -> faster-whisper (speech-to-text) -> Ollama llama3.2:3b (local LLM, token streaming) -> Piper (neural text-to-speech) -> speaker. The reply is spoken sentence-by-sentence while the LLM is still generating; measured mean latency to first spoken word is ~1.3-2 s. If the first sentence is not ready within 1.2 s, a rotating pre-recorded filler phrase keeps the user engaged; if the LLM is unreachable, the assistant answers with conversational recovery phrases - never a generic error message. The user can barge in (speak over the assistant) to interrupt and ask something else.